



INORGANIC FERTILIZER APPLICATION FOR COCONUT

Introduction

Coconut palm requires a continuous supply of plant nutrients for its high nut production. Supply of plant nutrients to palms is referred as fertilizer application. Plant nutrients are removed from a palm in the form of nuts, husks, fronds, and other materials in the crown

The approximate rates of major plant nutrients removed from palms in soil suitability classes S_1 and S_2 giving an average nut yield of 75 - 100 nuts per palm per year shown below.

Nitrogen	-	740 g (Urea equivalent of 1.6 kg)
Phosphorus	-	118 g (Eppawela Rock Phosphate equivalent of 0.9 kg)
Potassium	-	1553 g (Muriate of potash equivalent of 3.1 kg)
Magnesium	-	213 g (Dolomite equivalent of 1.7 kg)

The approximate rate of major plant nutrients removed from palms in S_4 and S_5 soil suitability classes giving an annual nut yield of 50 - 75 per palm is as follows.

Nitrogen	-	424 g (Urea equivalent of 0.9 kg)
Phosphorus	-	95 g (Eppawela Rock Phosphate equivalent of 0.7 kg)
Potassium	-	1248 g (Muriate of potash equivalent of 2.5 kg)
Magnesium	-	143 g (Dolomite equivalent of 1.2 kg)

In addition to the removal by palms as above, plant nutrients in the soil also deplete due to run off, leaching, erosion, and uptake by weeds etc

The above processes lead to depletion of essential plant nutrients in the soil, which cause a gradual decline of nut yield. Therefore, soils in coconut plantations should be enriched with nitrogen, phosphorus, potassium and magnesium by regular application of fertilizer. Coconut palms can be applied even with organic manures. This document presents complete details on inorganic fertilizer application for coconut.

The nutrient requirement of young coconut palms (before flowering) is different from that of adult coconut palms (after flowering). Separate fertilizer

Two sources of phosphate fertilizers, namely Eppawela Rock Phosphate (ERP) and Triple Super Phosphate (TSP) are recommended for coconut palms. ERP is recommended only for the wet and intermediate zones because its soluble phosphate fraction in 2% citric acid is low. Its dissolution rate is very slow in neutral soils in the dry zone.

A map of Sri Lanka showing boundaries of wet, intermediate and dry zones are shown in picture.



1. Straight fertilizers

Type of fertilizer	Nutrient and its percentages		
Urea	Nitrogen		46%
Eppawala Rock Phosphate	Phosphorus Pentoxide	(P ₂ O ₅)	28-30%
Triple Supper Phosphate	Phosphorus Pentoxide	(P ₂ O ₅)	46%
Muriate of Potash	Potassium Oxide	(K ₂ O)	60%
Dolomite	Magnesium Oxide	(MgO)	20%

1.1 Basal dressing per planting hole at the time of planting.

Wet and intermediate zones		Dry zone	
Type of fertilizer	Amount (g)	Type of fertilizer	Amount (g)
Urea	250	Urea	250
Eppawala Rock Phosphate	750	Triple Supper Phosphate	350
Muriate of Potash	250	Muriate of Potash	250
Dolomite	1000	Dolomite	1000

In addition to the inorganic fertilizer recommended above, application of 10 kg of dry cattle manure or goat manure or seasoned poultry manure (heaped in the field for about 2 -3 months) per planting hole would be beneficial.

1.2. For coconut seedlings and young palms

YPM requirements per palm in grams based on the age

1.2.1 For wet and intermediate zones

Age of palm →	6 months	1 year	1 ½ years	2 years	2 ½ years	3 years	3 ½ years	4 years up to bearing
Type of fertilizer ↓	fertilizer (g)							
Urea	190	235	235	305	305	375	375	470
Eppawela Rock Phosphate	420	530	530	690	690	850	850	1060
Muriate of Potash	190	235	235	305	305	375	375	470
Dolomite	500	500	500	500	500	500	500	500

1.2.2. For dry zone

Age of palm →	6 months	1 year	1 ½ years	2 years	2 ½ years	3 years	3 ½ years	4 years up to bearing
Type of fertilizer ↓	fertilizer (g)							
Urea	190	235	235	305	305	375	375	470
Triple Supper Phosphate	160	200	200	300	300	360	360	400
Muriate of Potash	190	235	235	305	305	375	375	470
Dolomite	500	500	500	500	500	500	500	500

1.3 For adult palms

Wet and intermediate zones		Dry zone	
Type of fertilizer	Rate (g)	Type of fertilizer	Rate (g)
Urea	800	Urea	800
Eppawela Rock Phosphate	900	Triple supper Phosphate	400
Muriate of Potash	1600	Muriate of Potash	1600
Dolomite	1000	Dolomite	1000

1.4 For improved cultivars

CRIC60, CRIC65 and CRISL98 and high yielding palms and high yielding palms (more than 75 nuts /palm/yr), application of 1 ½ times the above rates is recommended.

1.5 King Coconut Palms

Fertilizer recommendation to King Coconut is the same as that given in sections 1.1, 1.2 and 1.3.

1.6 Toddy Tapping Palms

Wet and intermediate zones		Dry zone	
Type of fertilizer	Rate (g)	Type of fertilizer	Rate (g)
Urea	1500	Urea	1500
Eppawela Rock Phosphate	2100	Triple super Phosphate	950
Muriate of Potash	4800	Muriate of Potash	4800
Dolomite	750	Dolomite	750

2. Fertilizer mixtures

2.1 NPK fertilizer mixture for coconut

2.1.1 Fertilizer mixture for basal dressings, seedlings and young palms

(a) Young Palm Mixture for the wet and intermediate zones (YPM-W)

Urea	2.0 parts by weight
Eppawela Rock Phosphate	4.5 parts by weight
Muriate of Potash	2.0 parts by weight

NPK composition of the mixture

Nitrogen (N) 11%, Phosphorus Pentoxide (P_2O_5) 16%, Potassium Oxide (K_2O) 14%, (11-16-14)

(b) Young Palm Mixture for the dry zone (YPM-D)

Urea	2 parts by weight
Triple Super Phosphate	2 parts by weight
Muriate of Potash	2 parts by weight

NPK composition of the mixture

Nitrogen (N) 13%, Phosphorus Pentoxide (P_2O_5) 15%, Potassium Oxide (K_2O) 20%, (13-15-20)

2.1.2 Fertilizer mixtures for adult palms

(a) Adult Palm Mixture for the wet and the intermediate zones – (APM-W)

Urea	8 parts by weight
Eppawela Rock Phosphate	9 parts by weight
Muriate of Potash	16 parts by weight

NPK composition of the mixture

Nitrogen (N) 11%, Phosphorus Pentoxide (P_2O_5) 8%, Potassium Oxide (K_2O) 29%, (11-8-29)

(b) Adult Palm Mixture for the wet and the intermediate zones – (APM-D)

Urea	8 parts by weight
Triple Super Phosphate	4 parts by weight
Muriate of Potash	16 parts by weight

NPK composition of the mixture

Nitrogen (N) 13%, Phosphorus Pentoxide (P_2O_5) 7%, Potassium Oxide (K_2O) 34%, (13-7-34)

2.2 NPK mixtures for King Coconut Palms

Fertilizer mixtures recommended for coconut under the section 2.1 are also recommended for King Coconut Palms.

Note:

- The APM fertilizers become moist when it is exposed to air, because its urea component absorbs moisture from the atmosphere. When the mixture is wet, it is difficult either to store or apply to the soil.
- APM should be applied when the soil surface is wet to minimize ammonia emission from urea.
- Although dolomite should be applied to the coconut palm as a magnesium fertilizer, it is not included into the APM mixtures because urea and dolomite should not be mixed and stored. However, it is not necessary to have a time gap between APM and dolomite application. APM and dolomite may be broadcast on the manure circle and incorporated with the soil in the same time.

3. Application rates of fertilizer mixtures

3.1 Basal dressing at planting

For each planting hole, the topsoil used for filling should be mixed with the following.

For wet and intermediate zone

YPM-W 1250g

Dolomite 1000g

For dry zone

YPM-D 850g

Dolomite 1000g

In addition to the inorganic fertilizer recommended above, application of one of the organic manures such as dry cattle manure, goat manure, seasoned poultry manure 10 kg per planting hole would be beneficial.

3.2 Application rate for young palms: Young Palm Mixture (YPM-W and YPM-D)

Age of seedling	Rate of application g/palm/ 6 months		
	YPM-W	YPM-D (Dry zone)	Dolomite
6 months	800	540	500
12 months (1 year)	1000	670	500
18 months (1 ½ years)	1000	670	500
24 months (2 years)	1300	905	500
30 months (2 ½ years)	1300	905	500
36 months (3 years)	1600	1110	500
42 months (3 ½ years)	1600	1110	500
48 months (4 years) up to bearing	2000	1340	500

3.3 Rate of application for adult palms

Adult palm mixture (APM - W, APM-D) and Dolomite

Rate of application kg/palm/yr		
APM-W (Wet & intermediate zones)	APM-D (Dry zone)	Dolomite
3.3	2.8	1.0

The above - recommended dosage should be applied for palms yielding 50-75 nuts per palm per year. For hybrid coconut, improved cultivars and high yielding palms (more than 75 nuts/palm/yr) application of 1.5 times the above rates is recommended.

3.4 King Coconut

Recommendations for King Coconut are the same as given in section 3.1,3.2 and 3.3.

4. Method of fertilizer application

4.1 Young palms

In the early stage (seedlings up to 1-1 ½ years) fertilizers should be broadcast close to the palm covering the entire circle (manure circle) up to a distance of 30 cm (about 1') from its base, and incorporate it well with the soil. As the palm grows older, this area should be gradually extended ($\frac{1}{3}$ m per year) up to about 1.8 m at the time of flowering (See the table given below). Apply a mulch dried coconut fronds or coconut husks immediately after fertilizer application (Refer 4.3). To obtain the full benefits of fertilizer application all the above steps should be followed.



figure 2: Manure circle with 6 feet radius

Age of the seedling	Radius of the manure circle (cm)
6 months	30
One year	60
1 ½ years	60
Two years	90
2 ½ years	90
Three years	120
3 ½ years	120
From 4 years up to flowering	150

4.2 Bearing palms

Fertilizer should be broadcast in the entire area of the soil surface around the base of the palm (manure circle) up to a distance of 1.8 meters (about 6') from the base. Then the fertilizer should be well mixed with the soil either with a mamoty or fork. (figures 3,4 and 5) Then the entire manure circle should be mulched well. (figure 5) If the weed growth of the manure circle is high it should be slashed (avoid clean weeding in the manure circle with a mamoty). All the above steps should be followed to obtain the full benefits of the fertilizer application.

4.3 Mulching

Immediately after fertilizer application, the manure circle should be completely covered with materials such as weed trash, dried fronds or husks (figure 5). Mulch helps to suppress ammonia emission from urea, suppress weed growth and conserve soil moisture in the manure circle. It also reduces soil temperature, prevents drying of the soil surface and increase organic matter content in the soil. If fertilizer application is not done at the correct time, the full benefits of fertilizer application cannot be obtained. On very steep lands that are subjected to considerable surface run-off, fertilizer should be application in a half circle shallow basin (up to 10 cm depth) cut round the palm on the upper part of the slope. The diameters of the inner circle and the outer circle should be 0.5 m and 1.5m respectively. After the fertilizer application in the basin it should be closed well and mulched.



figure 3: Spreading of fertilizer on manure circle



figure 4: Mixing of fertilizer with soil



figure 5: Mulching of manure circle

5. Frequency of application

5.1 Young palms

At least, half yearly application of fertilizer is recommended. When circumstances permit, 03 months split application may be adopted particularly for lands with sandy soils to minimize leaching of fertilizer due to heavy rainfall during monsoons. For 03 months split application, apply the half of the recommended dose given in Section 2.2

5.2 Bearing palms

Fertilizer application should be carried out annually. In areas where heavy rains are received during both monsoons and for lands with sandy soils, half yearly application with each monsoon may be adopted to minimize leaching losses. For half yearly application, apply half of the recommended doses given in section 1.3 and 1.4.

6 Differential Fertilizer Recommendation (DFR)

This is also known as the fertilizer recommendation based on leaf analysis. Differential Fertilizer Recommendation provides information regarding the soil nutrient deficiencies, hidden nutrient deficiencies and excess nutrient levels of palms. In this method, nutrient deficiencies are identified with leaf analysis and precise fertilizer recommendations are made by adding excess nutrients to correct deficiencies. This method is important to reduce unnecessary waste of fertilizers and help to formulate the most suitable fertilizer mixture for palms. It is important to consider that requests for DFR should be processed minimum of 6 months after the fertilizer application. The best months for taking leaf samples for DFR are January, February, June, July and August. Leaf samples for DFR cannot be taken during rainy seasons or within 6 months after fertilizer application. Growers can obtain details regarding leaf analysis services by making a request to the Head, Soils and Plant Nutrition Division of Coconut Research Institute.

7. Fertilizer application for irrigating palms

If coconut palms are irrigated, it is advisable to apply the recommended fertilizer in split doses with the supply of water. The water soluble fertilizers such as Urea and White Muriate of Potash could be applied with water while insoluble fertilizer such as Eppawala Rock Phosphate, Imported Rock Phosphate and Dolomite are applied separately to the manure circle.

Soils and Plant Nutrition Division
January 2018